Weisheng Zhang

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Research Fields
Econometrics, particularly network and panel data.

Education

University College London — London, UK
Ph.D. in Economics — May 2027 (expected)
Advisors: Aureo de Paula and Andrei Zeleneev
MRes in Economics — 2022

University of Notre Dame — IN, US
Master's in Economics — 2021

Peking University — Beijing, China
Master's in Economics — 2020
B.S. in Geophysics — 2017

Job Market Paper
"*Causal Inference under Dynamic Selection and Latent Heterogeneity*" 2026

Abstract: I study dynamic treatment effects in panel data under staggered adoption when treatment timing depends jointly on unobserved time-invariant heterogeneity and time-varying pretreatment covariates, including lagged outcomes. Untreated potential outcomes follow a nonparametric dynamic panel model that allows flexible interactions between time-varying covariates and latent heterogeneity. I use pretreatment outcome histories to find individuals with similar time-invariant latent factors, and the key requirement is that these histories are sufficiently informative about those latent factors. I develop an identification strategy for the dynamic average treatment effect on the treated (ATT) and propose kernel-based doubly robust estimators for the dynamic ATT. I further combine double cross-fitting with undersmoothing and show that, under suitable regularity conditions, the proposed estimators are root-n consistent, asymptotically normal, and asymptotically unbiased. The simulation study demonstrates that the proposed method provides accurate inference across a wide range of data-generating processes. I illustrate the method with an application to the U.S. family planning program studied by Bailey (2012) and reestimate its effects on fertility rates.

Working Papers

1. "*Tractable Estimation of Nonlinear Panels with Interactive Fixed Effects*" (with Andrei Zeleneev) [Submitted] 2025
   Abstract: Interactive fixed effects are routinely controlled for in linear panel models. While an analogous fixed effects (FE) estimator for nonlinear models has been available in the literature (Chen, Fernández-Val and Weidner, 2021), it sees much more limited use in applied research because its implementation involves solving a high-dimensional non-

convex problem. In this paper, we complement the theoretical analysis of the FE estimator by providing a new computationally efficient estimator that is asymptotically equivalent to the FE estimator. Unlike the previously proposed FE estimator, our estimator avoids solving a high-dimensional non-convex optimization problem and can be feasibly computed in large nonlinear panels. Our proposed method involves two steps. In the first step, we convexify the optimization problem using nuclear norm regularization (NNR) and obtain preliminary NNR estimators of the parameters, including the fixed effects. Then, we find the global solution of the original optimization problem using a standard gradient descent method initialized at these preliminary estimates. To make our method readily applicable in practice, we also propose specific numerical algorithms for solving the involved optimization problems, establish their convergence, and provide their efficient implementation in our R package *NNRPanel*.

2. "*Flexible Imputation of Incomplete Network Data*" (with Ge Sun) 2026
[Submitted]
Abstract: Sampled network data are widely used in empirical research because collecting complete network information is costly. However, empirical analyses based on sampled networks may lead to biased estimators. We propose a nonparametric imputation method for sampled networks and show that empirical analyses based on imputed networks yield consistent estimates. Our approach imputes missing network links by combining a projection onto covariates with a local two-way fixed-effects regression. The method avoids parametric assumptions, does not rely on low-rank restrictions, and flexibly accommodates both observed covariates and unobserved heterogeneity. We establish entrywise convergence rates for the imputed matrix and prove the consistency of generalized method of moments (GMM) estimators based on imputed networks. We further derive the convergence rate of the corresponding estimator in the linear-in-means peer-effects model. Simulations show strong performance of our method in both imputation accuracy and downstream empirical analysis. We illustrate our method with an application to the microfinance network data of Banerjee et al. (2013).

Work in Progress

1. "*Panel Models with Interactive Fixed Effects and Low-Rank Regressors*"
Abstract: A novel method is developed for identification and estimation in linear panel models with interactive fixed effects and low-rank regressors. Rather than partialing out the low-rank component of the regressors, the proposed method directly exploits their low-rank structure through a rank-reduction criterion.

2. "*An Econometric Analysis of Non-Bayesian Social Learning*" (with Shuyang Sheng, Guanyi Wang, and Yuanqi Zhang)
Abstract: We are developing an econometric framework for studying non-Bayesian social learning in networks, with an empirical application to agricultural technology adoption. Our analysis combines farmer-level data with a microfounded model in which farmers update their beliefs about a new technology using their own and their neighbors' harvest outcomes and make adoption decisions based on these beliefs.

Teaching Experience

ECON0019, Econometrics (UCL, Undergraduate)	2025 - Present
ECON0010, Math for Economists (UCL, Undergraduate)	2024 – 2025
ECON0115, Household Economics (UCL, Undergraduate)	2022 - 2024
Chinese Economy (Peking University, Undergraduate)	2018

Seminar and Conference Presentations

2026: Econometric Society East & Southeast Asia Meeting (Hanoi), Econometric Society Africa Meeting (Cairo), UCL-CeMMAP Ph.D. Econometrics Research Day, UCL Econometrics Brownbag Seminar.
2024-25: UCL-CeMMAP Ph.D. Econometrics Research Day, UCL Econometrics Brownbag Seminar.

Scholarship

Centre for Microdata Methods and Practice (CeMMAP) PhD scholar, 2022-present

Skills and Languages

Languages: English (fluent), Mandarin Chinese (native)
Programming: Python, R, C++

References

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Last updated: Oct 26, 2026